

Assignment - 02

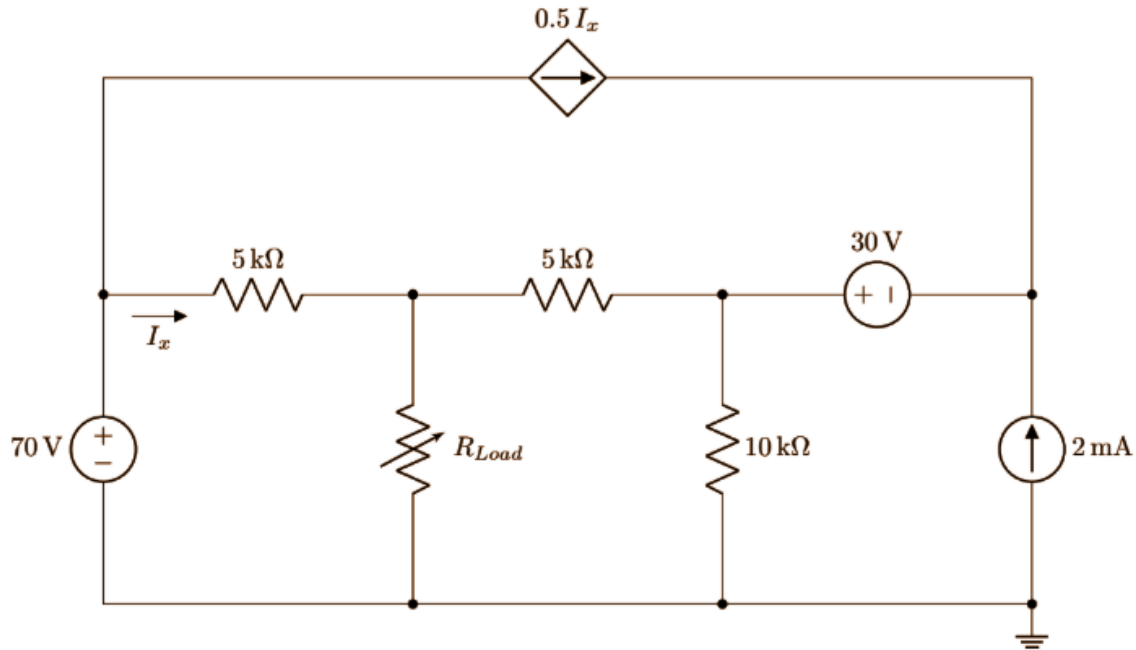
Department of CSE, BRAC University

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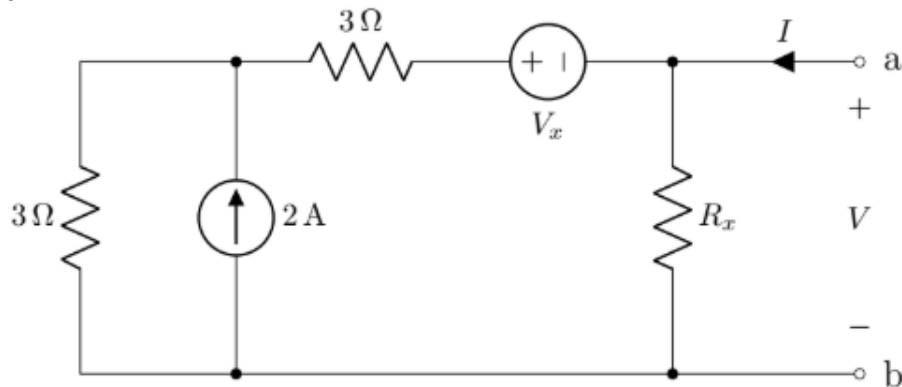
Course No: CSE250

Total Marks: 5

- 1) Apply Thevenin's / Norton's Theorem to find the value of R_{Load} for which the circuit will deliver maximum power to it. Also, what is the maximum power that can be delivered to the load?

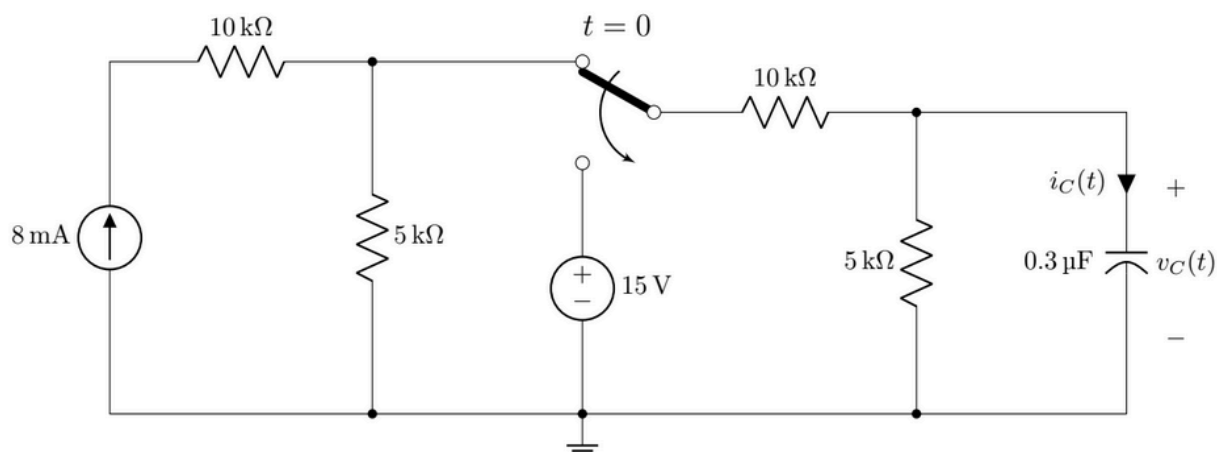


- 2) In the circuit below, all resistors are practical and cannot have negative values. When a voltage of $V = 2\text{ V}$ is applied across terminals 'a' and 'b', the circuit draws a current of $I = 3\text{ A}$. However, if -6 V is applied, instead of drawing, the circuit supplies 1 A to the terminal 'a'.

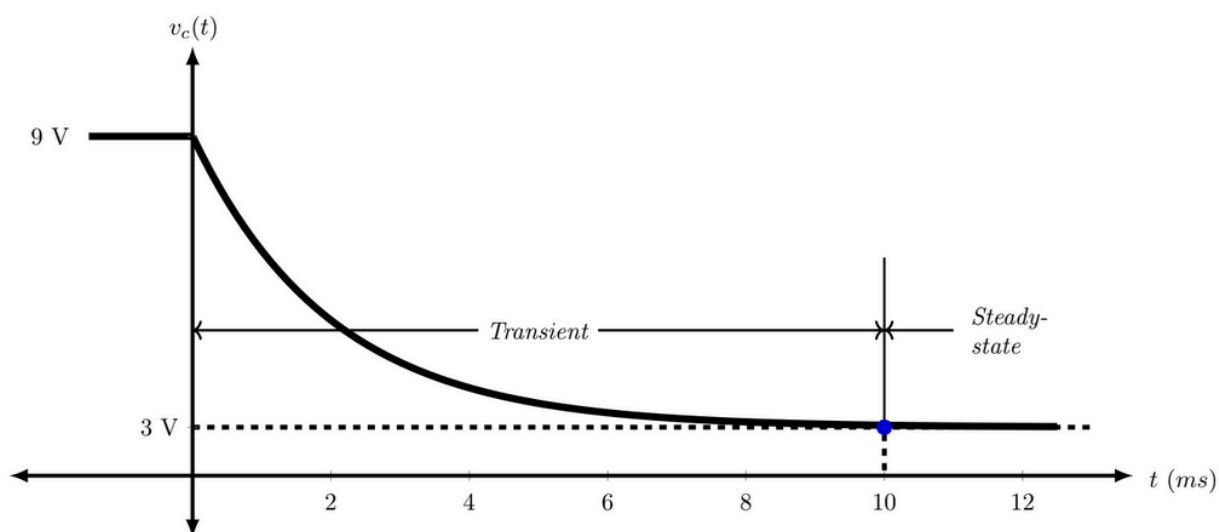


Determine the unknown resistance R_x in the circuit. Also, find an equivalent representation of the circuit having a minimum number of elements and determine V_x .

- 3) The switch in the following circuit shifts at $t = 0$. Determine the voltage response of the capacitor $V_c(t)$ and current through the capacitor $i_c(t)$ as a function of time for $t > 0$.

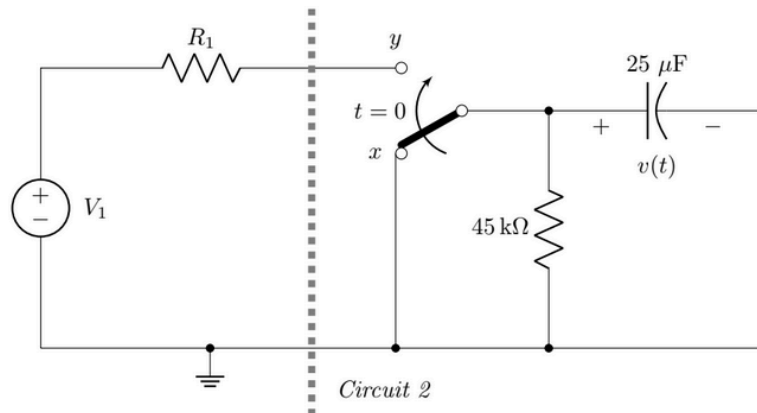
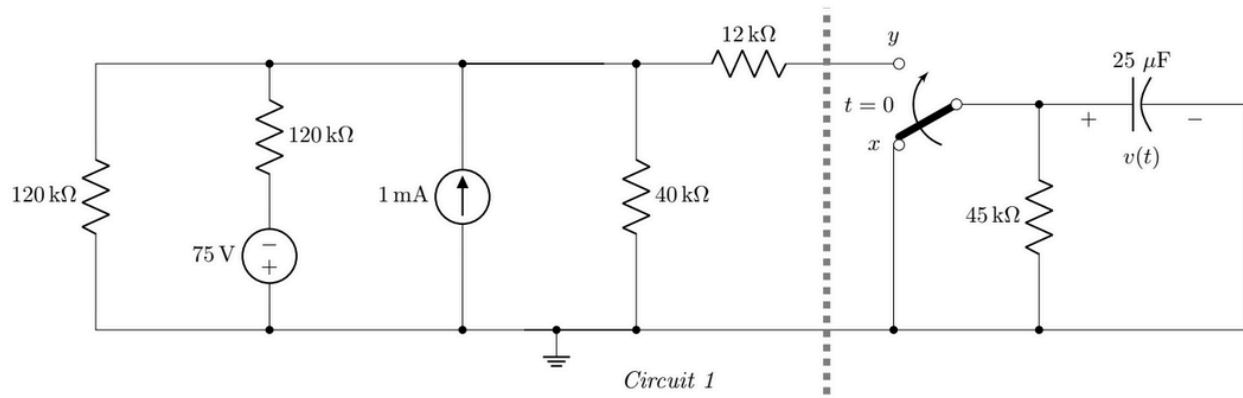


- 4) The $V_C(t)$ vs t plot below shows the voltage response of a capacitor (C) in a series RC circuit to a sudden change in the DC voltage applied through an equivalent resistance of $2\text{ k}\Omega$.



- Determine the approximate Time Constant from the figure. Determine capacitance C with the appropriate unit.
- Write a mathematical expression $V_C(t)$ for $t > 0$.
- Predict and draw a circuit with an appropriate switching mechanism that can generate the voltage response as shown in the plot.

5)



- A. Apply Source Transformation to reduce the left portion with respect to the dashed gray line of Circuit 1 so that it takes the form of Circuit 2 as shown. Write down the values of V_1 and R_1 .
- B. Then, analyze the Transient Behavior of circuit 2, assuming that the switch moves from position x to position y at $t = 0$. Determine $V(t)$ for $t > 0$.